

AGENT EVALUATION LAB

CAPSTONE PROJECT

AGENDA

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- Problem Statement
- Project Description
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- Technologies & Algorithms
- Team Roles & Working Agreement
- Sprint Plan & Schedule
- Retrospective

PROBLEM STATEMENT

AI agents such as large language models (LLMs), Retrieval-Augmented Generation (RAG) systems, and conversational assistants are increasingly being deployed in real-world applications like customer support, healthcare, education, and enterprise decision-making.

However, these AI systems often face serious challenges, including:

- Unsafe or harmful responses
- Vulnerability to jailbreak and prompt injection attacks
- Hallucinated or incorrect outputs in RAG-based systems
- Lack of consistent reliability and behavioral evaluation

Currently, most AI evaluation processes are manual, inconsistent, and difficult to scale, making it hard for developers and organizations to ensure AI safety and robustness before deployment.

Therefore, there is a need for an automated, systematic, and extensible framework that can evaluate and red-team AI agents across safety, reliability, and behavioral scenarios, while providing clear scoring and actionable reporting.



PROJECT DESCRIPTION

AGENT EVALUATION LAB IS A PRODUCTION-GRADE FRAMEWORK THAT:

- AUTOMATICALLY TESTS AI AGENTS
- DETECTS SAFETY VIOLATIONS AND JAILBREAK ATTACKS
- EVALUATES RELIABILITY AND HALLUCINATIONS
- GENERATES PASS/FAIL REPORTS

SUPPORTS:

- OPENAI MODELS
- RAG SYSTEMS
- CUSTOM AI SERVICES

THIS IS AN ENTERPRISE-READY EVALUATION PLATFORM.

Team Member Roles & Responsibilities

- Sreenitha Rayapuraju – System Architecture & Evaluation Engine Design
- Sai Teja Kusireddy– Backend Development & OpenAI/RAG Adapter Integration
- Koushika Chappidi – Scenario Creation & Automated Testing (YAML + PyTest)
- Veera Sai Akshitha Punyamanthula– Reporting System & Dashboard Visualization
- Harshitha Reddy Mannemala– DevOps, CI/CD Pipeline & Documentation

PERSONAS

Persona 1: AI Application Developer

Name: Alex Johnson

Role: Machine Learning Engineer / AI Developer

Background: Builds AI-powered chatbots and assistants using OpenAI APIs.

Goals:

- Ensure the AI agent behaves correctly before deployment
- Test safety and reliability automatically
- Detect jailbreak or prompt injection vulnerabilities early

Pain Points:

- Manual testing is slow and inconsistent
- Difficult to track failures across multiple scenarios
- Needs structured evaluation results for debugging

How Agent Evaluation Lab Helps:

Provides automated scenario-based testing with clear PASS/FAIL scoring and detailed reports.

Persona 2: Enterprise AI Safety & Compliance Engineer

Name: Kate Sharma

Role: AI Safety Specialist / Compliance Officer

Background: Responsible for ensuring AI systems meet enterprise safety and ethical standards.

Goals:

- Prevent unsafe or harmful AI outputs
- Ensure compliance with organizational policies
- Generate audit-ready safety evaluation reports

Pain Points:

- Hard to validate AI safety across many use cases
- Lack of standardized red-teaming tools
- Needs clear documentation of risks and failures

How Agent Evaluation Lab Helps:

Runs red-team safety scenarios and produces professional reports in Markdown/HTML/JSON.

Persona 3: RAG System Engineer

Name: Anthony

Role: Software Engineer (RAG & Search Systems)

Background: Develops Retrieval-Augmented Generation (RAG) applications for enterprise knowledge assistants.

Goals:

- Ensure responses stay grounded in retrieved documents
- Detect hallucinations and missing-context failures
- Improve reliability of enterprise AI assistants

Pain Points:

- RAG systems often hallucinate when context is incomplete
- Difficult to evaluate retrieval + generation quality together
- Needs automated reliability checks

How Agent Evaluation Lab Helps:

Includes RAG-specific scenarios such as missing-context tests and hallucination detection.

TECHNOLOGIES

Technologies Used

- Python 3.11
- OpenAI API
- YAML Scenario Files
- GitHub Actions (CI/CD)
- PyTest for testing
- Ruff for linting
- Markdown / HTML /JSON reporting

Optional Future Tools:

- Docker
- FastAPI for RAG demo

ALGORITHMS

Algorithms / Methods Used

- Rule-based evaluation predicates
- Prompt-injection detection scenarios
- Reliability scoring metrics
- Hallucination detection checks

Future additions:

- ML-based evaluation scoring
- Reinforcement Learning evaluation
- Statistical confidence scoring

SYSTEM ARCHITECTURE

- Agent Evaluation Lab uses a modular pipeline to test AI agents using structured scenarios and automated scoring.
- **Inputs:**
 - CLI command starts an evaluation run
 - config.yaml selects provider/model + runtime settings
 - YAML scenario files define prompts, tags, retries, and PASS/FAIL criteria
- **Core Execution:**
 - Scenario Loader reads YAML scenarios
 - Runner executes scenarios (supports retries + concurrency)
 - Evaluator applies predicate-based checks → PASS / FAIL_MINOR / FAIL_CRITICAL
 - Failure reasons are logged for unsafe or incorrect behavior
- **Adapters (Agent Connectivity):**
 - OpenAI Adapter → evaluates OpenAI models
 - HTTP Agent Adapter → evaluates deployed agents (RAG/custom APIs)
- **Outputs:**
 - Markdown report (human-readable summary)
 - HTML report (visual/interactive)
 - JSON report (CI/CD automation + metrics)

SPRINT 0 COMPLETED TASKS

During Sprint 0, our team completed:

- Finalized project idea and scope
- Created GitHub repository and enabled Wiki
- Defined project architecture and workflow
- Selected tools and programming stack
- Prepared Sprint 0 documentation pages
- Drafted team working agreement
- Planned Sprint 1 development tasks
- Sprint 0 establishes the foundation for implementation.

TEAM WORKING AGREEMENT

Our team agreed on:

- Weekly meetings every Thursday
- Communication via WhatsApp and GitHub Issues
- Tasks must be committed at least 24 hours before deadlines
- Equal contribution from all members
- No last-minute work dumping
- Each member is responsible for completing assigned tasks independently

SPRINT 1 PLAN

- In Sprint 1, we will focus on implementation.
- Build the core scenario execution engine.
- Add initial safety and jailbreak evaluation scenarios.
- Run evaluations on sample AI agents.
- Generate the first automated reports.
- Begin Sprint burndown tracking with completed tasks.
- Prepare for Sprint 1 review presentation.

EXPECTED DELIVERABLES

- By the end of the project, we aim to deliver:
- Fully working AI evaluation framework
- Scenario library for safety and reliability testing
- Report generation system
- CI/CD workflow for continuous evaluation
- Complete GitHub documentation and wiki
- Final presentation and demonstration

The background features a minimalist design with several thin, light-colored lines intersecting across the frame. A large, light-colored triangle is positioned on the right side, pointing towards the center. The text "THANK YOU" is centered in a bold, dark serif font.

THANK YOU